Undeniability Does Not Yield Succession

Abstract

A tokening that marks the denial of occurrence differently from its assertion witnesses occurrence. That retorsive step is propositional once the tokening fact is an antecedent, and it is formalizable over an arbitrary domain of loci. Temporal succession is another matter. If claims are properties of succession-free worlds, then two tensed structures over one world satisfy exactly the same claims; in particular, a still structure and a structure with nonempty succession are claim-indiscernible. Consequently, no undeniable claim — no claim true at every world that discriminates its denial — can entail running. Relative to the signature (T, h, tok), an oriented succession premise P is independent of every claim true at a shared two-model witness: no such claim entails P for all tensed worlds, and none refutes it. On loci with decidable equality, moreover, every relabeling-invariant succession is symmetric, so orientation is not available from invariance. Structural consequences of a bare distinction (two-valued arena, uniqueness of complement, fixed-point freeness, period-two alternation) are available as verified conditionals; none of them yields succession. The results are checked in Lean 4 with an empty axiom footprint. The upshot is a limit theorem for retorsive arguments: discrimination reaches occurrence; it does not reach time.

1 Introduction

Retorsive and transcendental arguments aim to extract objective content from the conditions of assertion or denial [26, 25]. A standing worry is that such arguments report only on what a denier must accept, leaving the world untouched. The present paper isolates one place where the worry is decisive and one place where it is not.

The positive half is familiar in outline. Let a medium classify propositions by assigning each a locus and reading loci with a two-valued map. If the denial of occurrence and its assertion receive different readings, then occurrence holds: the two loci witness a distinction. Once the tokening difference is an antecedent, the result is a propositional refutation of the conjunction of that difference with the denial [15, 10]. Section 2 states the theorem.

The negative half is the contribution. Regiment claims as properties of *worlds* that carry loci, a reading, and a tokening, but no succession relation. A tensed world adds succession. Under that typing, succession is invisible to claims: any two tensed structures over one world satisfy exactly the same claims (Section 3). It follows that no claim true wherever its denial is discriminated can entail running. Temporal realization therefore requires a premise P outside the undeniable fragment. Relative to the world signature, P is independent of every claim true at the two-model witness of Section 5. A further theorem locates the directional content of P: on loci with decidable equality, invariant succession is always symmetric (Section 6).

The structural facts about a two-valued carrier — that some equivariant law moves a point exactly when the carrier has two elements, that the excluding structureless law is complement, that complement has no fixed point and yields a unique period-two orbit — are recorded in Section 7 as background, with proofs in the accompanying Lean development. They are not used to derive succession. The paper's thesis is the limit, not an existence proof for becoming.

Section 8 answers the objection that claims ought to be tensed. Section 9 situates the result. Section 10 records the verification.

2 Discrimination

Definition 2.1 (Occurrence)**.** *For $h : T \to \mathbb{B}$, write* $\mathrm{Occurs}(h)$ *when there exist $t_1, t_2 \in T$ with $h(t_1) \neq h(t_2)$.*

Theorem 2.2 (Discrimination)**.** *Let* $\mathrm{tok} : \mathrm{Prop} \to T$ *and $h : T \to \mathbb{B}$. If*

$$h(\mathrm{tok}(\neg\mathrm{Occurs}(h))) \neq h(\mathrm{tok}(\mathrm{Occurs}(h))),$$

then $\mathrm{Occurs}(h)$ *and the two tokens are distinct.*

Proof. The two values of $h \circ \mathrm{tok}$ are distinct elements of the image of h, so $\mathrm{Occurs}(h)$ holds. If the tokens were equal, the values of h at them would be equal. □

The argument is cardinality under a self-referential instance: any tokening that takes two values shows that the medium has two elements; aiming the pair at ¬Occurs and Occurs refutes the denial. In Mackie's taxonomy the result is absolute (propositional) once the tokening fact is an antecedent [15]. It is the explicit-tokening form of Hintikka's existential inconsistency [10].

The antecedent is not a theorem of the ambient type theory. A closed normal term of type $\mathrm{Prop} \to \mathbb{B}$ cannot depend on its argument, hence is constant. Detachment is ostensive: some actual classification separates the two readings. One concrete discharge uses the pair

> Occurrence fails: every reading of this page classifies alike.
> Occurrence holds: two readings of this page classify differently.

Any h that separates those two readings satisfies the hypothesis. The paper assumes such a discharge for the limit argument below; nothing later requires more than the conditional. In particular, the obstruction does not depend on which medium supplies the still discriminating world — only on there being one.

Remark 2.3. *On $\mathbb{B}$, distinctness coincides with being one complement apart. The third conjunct of the verified form of Theorem 2.2 is therefore redundant on two values. No temporal content is involved: a static pair of marks already satisfies the theorem. The formal alternation of Section 7 is likewise an ω-indexed sequence of values, a tenselessly specifiable object. Realization in succession is a separate question, taken up next.*

3 Worlds, freeze, and obstruction

Definition 3.1 (Worlds and claims)**.** *A* world *is a triple $W = (T, h, \mathrm{tok})$: a type of loci, a reading $h : T \to \mathbb{B}$, and a tokening* $\mathrm{tok} : \mathrm{Prop} \to T$*. A* claim *is a property of worlds. W* discriminates the denial *of a claim A when $h(\mathrm{tok}(\neg A(W))) \neq h(\mathrm{tok}(A(W)))$. A is* undeniable *when $A(W)$ holds at every world W that discriminates the denial of A.*

Definition 3.2 (Tensed worlds)**.** *A* tensed world *is a world together with a binary succession relation on its loci. It is* still *when the relation is empty. It* runs *when there is an injective sequence of loci, each succeeding the last, whose readings alternate under complement.*

Occurrence, read as a claim, is undeniable: Theorem 2.2 transposed to worlds. Undeniability is nevertheless weak.

Proposition 3.3 (Calibration)**.** *The constant-true claim is undeniable.*

Proof. The consequent is trivial at every world. □

So undeniability constrains a claim only when the discrimination hypothesis reaches the claim's content. Occurrence is such a claim; an arbitrary bridge to running need not be.

Proposition 3.4 (Freeze)**.** *For any world W and any two succession relations R_1, R_2 on its loci, every claim holds of the tensed world (W, R_1) if and only if it holds of (W, R_2). In particular, over any inhabited world a still structure and a structure with nonempty succession coexist and satisfy exactly the same claims.*

Proof. Claims are typed over worlds, not over tensed worlds. The two tensed worlds share W, so they satisfy the same claims by definition. Equip W with the empty relation and with the total relation for the second clause. □

Theorem 3.5 (Obstruction)**.** *Let A be undeniable and suppose A entails running. Then no still tensed world discriminates the denial of A.*

Proof. Suppose a still tensed world discriminates the denial of A. By undeniability, A holds at its underlying world. By the entailment, the tensed world runs. But a still world has no succeeding pair, so it does not run. □

Corollary 3.6 (No undeniable bridge)**.** *If some still tensed world discriminates the denial of A, then no undeniable A entails running.*

The corollary is a rebuttal schema against bridges to *running*. Given a candidate axiom offered as an undeniable entailment of Runs, instantiate a still world that discriminates its denial (e.g. inscriptions on a page with empty succession, once a separating reading is granted). The candidate fails. The schema does not close the gap for candidates that are never so tokened; it shows that undeniability, as defined, cannot be that bridge. A parallel observation for the ordering half of P is recorded in Section 4.

Two readings of the loci are useful. On the *act* reading, T comprises readings performed by a reader; on the *mark* reading, T comprises inscriptions. The still witness for Corollary 3.6 uses marks with empty succession. A reader who insists that the inscriptions were produced in order is describing a second tensed structure over the same world; by Proposition 3.4 the two agree on every claim.

4 The succession premise

Definition 4.1 (Oriented succession)**.** *Fix a tensed world and a tokening. The* oriented succession premise *at a proposition X is*

$$P(X): \quad \mathrm{succ}(\mathrm{tok}(\neg X), \mathrm{tok}(X)) \ \wedge\ \neg\mathrm{succ}(\mathrm{tok}(X), \mathrm{tok}(\neg X)).$$

The first conjunct is the ordering half; the second is the orientation half.

Over a fixed world, the empty succession makes every instance of P fail and the total succession makes every ordering-half instance hold (`premise_fails_still`, `premise_holds_running`); Proposition 3.4 ensures the two structures agree on all claims. Moreover, no undeniable claim can entail the ordering half of P: if it did, any still world discriminating that claim's denial would satisfy the claim and hence witness a succession, contradicting stillness. (Corollary 3.6 is the companion fact for entailments of Runs.) On the act reading, a single instance of P is an ordinary report of reading order: the denial was read, then the assertion, and not conversely [18]. Warrant for one instance does not automatically extend to an infinite family of instances.

Theorem 4.2 (One instance)**.** *Given discrimination at X and one instance of $P(X)$, the two tokens are distinct, the successor's value is the complement of the predecessor's, and the structure is not still.*

Proof. Distinctness follows from discrimination (equal tokens yield equal readings). The value claim is the two-element fact that distinct Boolean values are complements. Non-stillness is immediate from the ordering conjunct. □

Theorem 4.3 (Chain)**.** *A chain of oriented, stepwise-discriminated instances over pairwise-distinct loci yields an oriented run, hence a run, and refutes stillness.*

Proof. Assemble the sequence of tokens. Injectivity is the pairwise-distinctness hypothesis; each step contributes forward succession and failure of the reverse; values alternate by the two-element fact as in Theorem 4.2. Stillness fails at the first step. □

Theorem 4.2 uses only the ordering conjunct of P; the orientation conjunct is idle in its proof. Orientation is needed for distinctness without discrimination, and for oriented runs as opposed to mere runs. Symmetric adjacency on $\mathbb{N}$ — the relation $n\,R\,m \leftrightarrow |n - m| = 1$ — runs and is not still, yet admits no oriented run, since every edge is reversible.

Three commitments should be kept distinct:

(1) one oriented instance of P — locally easy to grant on the act reading;

(2) an infinite run — a further assumption, not discharged by (1);

(3) A-theoretic passage — a further gap beyond tenseless succession [29, 16].

The obstruction concerns the step from undeniability to (1)–(2). Item (3) is left open even if (2) is granted. The paper claims no derivation of (2) or (3) from retorsion.

5 Independence

Proposition 5.1 (Independence)**.** *There is a world with two distinctly read loci such that:*

(i) the still tensed structure over it makes P fail for every tokening;

(ii) a tensed structure over it makes the ordering half of P hold for every tokening;

(iii) every claim holds at both structures or at neither.

Hence no claim true at that shared world *entails P for all tensed worlds, and none such refutes P* (`no_claim_settles_premise`)*. The unrestricted reading is false: the constant-false claim vacuously entails and refutes every target. The same exhibition works for oriented P over a world whose tokening separates a chosen pair. The full chain hypothesis is realized over a world with infinitely many loci (total succession on $\mathbb{N}$ with alternating reading, or successor on $\mathbb{N}$ for the oriented run).*

Proof. Take $T = \mathbb{B}$, $h = \mathrm{id}$, and a constant tokening for the still/total pair; Proposition 3.4 gives (iii), and the P-behaviour is as above. For oriented independence, take succession to hold of $(\bot, \top)$ alone. For the chain hypothesis, take $T = \mathbb{N}$ with the total relation and alternating reading (`chain_hypothesis_realizable`); successor on $\mathbb{N}$ supplies the oriented run (`oriented_witness_runs`). □

Independence is relative to the signature (T, h, tok). If the signature is enriched with causal facts and causal order determines temporal order, P may become derivable [17]. The present signature is exactly what retorsion uses. Putnam's relativity argument concerns frame-independent succession between spacelike-separated events [21]; successive readings by one reader are timelike.

6 Orientation

Running constrains the supply of loci before it constrains succession: a one- or two-element carrier admits no injective infinite sequence, whatever the relation.

Call a succession relation *invariant* when it is preserved by every transposition of loci. (Loci are assumed to have decidable equality, so that transpositions are definable from equality.)

Proposition 6.1 (Classification)**.** *An invariant succession is uniform on the diagonal and, given two distinct loci, uniform off the diagonal: it is determined by the propositions $R(x,x)$ and $R(x,y)$ for any $x \neq y$. Equality and the empty relation carry no run. An invariant succession that fails somewhere on the diagonal and holds of some off-diagonal pair is disagreement. When the relation is decidable (equivalently, $\mathbb{B}$-valued), the diagonal and off-diagonal values each case into true or false, yielding exactly four extensionally possible relations: empty, equality, disagreement, and total.*

Proof sketch. One transposition carries any diagonal pair to any other, so the diagonal is uniform. Two transpositions carry any off-diagonal pair to any other when at least two loci exist, so the off-diagonal is uniform. The verified content is this biconditional uniformity (`invariant_succ_classified`); a four-way exhaustive classification for arbitrary Prop-valued relations would decide those two propositions and is not claimed. A succession contained in equality identifies consecutive terms of any sequence, against injectivity. The last claim: failure on the diagonal and success off it force $R(u,v) \leftrightarrow u \neq v$ by the two uniformities (`invariant_says_is_ne`). □

Theorem 6.2 (Invariant succession is symmetric)**.** *Every invariant succession is symmetric. Hence no invariant succession is asymmetric on any pair, and no invariant succession carries an oriented run.*

Proof. For $u = v$ symmetry is trivial. For $u \neq v$, off-diagonal uniformity (Proposition 6.1) yields $R(u,v) \leftrightarrow R(v,u)$. An oriented run requires a step whose reverse fails. □

Thus invariance determines the *shape* of succession and not its *direction*. Three witnesses make the separation concrete:

1. symmetric adjacency on $\mathbb{N}$ runs, is not still, and holds in both directions at once;
2. the total relation satisfies the ordering half of P for every tokening and the oriented half for none;
3. successor on $\mathbb{N}$ yields an oriented run.

A C-theorist who accepts order without direction [6, 16] is therefore stable under any invariance argument: symmetric adjacency is a binary proxy for betweenness. The orientation clause of P is a genuine addition to the ordering clause. What it secures is local irreversibility; a succession may satisfy oriented-run and still contain a directed cycle (successor on $\mathbb{N}$ plus one reverse edge). Global topology remains open.

7 Background structure

For completeness, the conditional structural results used only as background are summarized here. Proofs are in `lean/Becoming.lean`.

1. *Arena.* On a carrier with decidable equality, there exists an equivariant endofunction that moves some point if and only if the carrier has exactly two elements (`moves_iff_two`). For mark-respecting equivariance (commutation with transpositions fixing a distinguished mark m), the matching facts are: some mark-respecting map moves m iff the carrier has two elements, and some mark-respecting map is fixed-point-free iff the carrier has two elements (`mark_moves_iff_two`, `marked_refuses_iff_two`). Moving an unmarked point is weaker: the constant map to m is mark-respecting on any carrier and moves every $x \neq m$ (`marked_three_moves`).

2. *Uniqueness.* On $\mathbb{B}$, the structureless (complement-equivariant) unary laws are the identity and Boolean complement. Only complement excludes a diagonal entry.

3. *No fixed point; alternation.* Complement has no fixed point. Its orbit from either seed is the unique period-two solution of the recursion; the two orbits are relabelings. Fixed-point freeness is equivalent to stepwise movement of every orbit.

4. *Articulate relations.* Definable binary relations on $\mathbb{B}$ that exclude something and are satisfied (objectually or processually) are extensionally inequality; satisfaction is then necessarily processual.

Each item is a conditional or a classification. None mentions succession on loci. Detaching occurrence for an actual medium uses the ostension of Section 2; detaching a temporal run uses P.

8 The regimentation dilemma

The obstruction has the succession-blind typing of claims as an antecedent. An objector may reply that tokenings are acts, acts are events, and the relevant claims are therefore tensed [2, 10]. Grant the proposal. Discrimination at the tensed level then either uses only the world components or also uses the succession component.

Theorem 8.1 (First horn)**.** *Suppose tensed-level discrimination depends on the world components alone. Let A' hold at every tensed world whose world part discriminates, and let A' entail running. Then no still tensed world discriminates.*

Proof. As in Theorem 3.5, one level up. □

Theorem 8.2 (Second horn)**.** *Suppose tensed-level discrimination requires that the two tokens stand in succession. Then discrimination is an instance of the ordering half of P, and any predicate entailing a succession fact refutes stillness wherever it holds.*

Proof. The first claim is definitional. The second: a succeeding pair contradicts stillness. □

On the first horn the obstruction survives. On the second, temporal content is built into discrimination, so it is assumed rather than derived. Either way, undeniability alone does not produce succession. The same boundary appears in Kaplan's distinction between context-validity and necessity [12]: a sentence may hold whenever tokened and still fail to be necessary.

9 Related work

Transcendental arguments. Stroud's limit — that such arguments reach the denier's commitments rather than the world — is the template [26]. Theorem 2.2 passes the limit for occurrence: an objective distinction follows from a discrimination hypothesis. Theorem 3.5 reinstates the limit for succession. In Stern's terms the result is ambitious for occurrence and modest for succession [25]. Illies's defence of ambitious transcendental arguments in moral philosophy is among the targets of the obstruction if those arguments require temporal or dynamical import from undeniability alone [11].

Final justification. Albert's Münchhausen trilemma lists regress, circularity, and break-off [1]. Apel and Kuhlmann seek to avoid break-off by performative self-defeat [2, 14]; Habermas declines strong ultimate grounding [8]. The present position accepts break-off for the temporal case: the conditionals are proved, P is independent of the claim language, and the obstruction explains why an undeniable temporal bridge is unavailable. The acceptance is local to succession; occurrence remains on the ambitious side.

Hegel and Trendelenburg. Trendelenburg charged that Hegel's derivation of movement imports intuition [27, 9]. Theorem 3.5 is a formal analogue under a succession-blind claim language: such a language entails no succession. Section 8 defends the regimentation by showing that the tensed alternative reproduces the same accounting. The paper does not offer a reading of the Logic; it records that a presuppositionless combinatorial beginning, if regimented as here, stops short of succession.

Temporal ontology. C-theory keeps order without direction [16, 6]. Theorem 6.2 shows that refusal of direction is stable under relabeling. The formal run, even when granted, is at-at variation along a succession, not A-theoretic passage [29, 23, 20, 22]. Mellor's token-reflexive account matches the tok/h signature and imposes the causal scope condition of Section 5 [17]. Grünbaum's mind-dependent becoming places temporal content in awareness of events [7], adjacent to a tokening-based account but silent on the obstruction.

Laws of Form. Spencer-Brown's oscillating value [24] and related three-valued or eigenform proposals [28, 13] address the structural half of Section 7. The present paper does not re-derive that half; it separates it from succession and assigns the latter to P. Under mark-respecting equivariance, a third value does not yield mark-moving dynamics (Section 7, item 1).

Verified arguments. Machine-checked ontological arguments typically place existential content in axioms and verify the derivation [19, 3, 4]. Here no axiom is declared. Existential temporal content sits in P and in the ostensive discharge of discrimination; the verified material is the surrounding conditional and limit structure.

10 Verification

The definitions and theorems of Sections 2–8 are named declarations in two standalone Lean 4 files [5], prelude only, with no imports and no axioms. Each file ends with a `#print axioms` audit; every line reports independence from all axioms, including Lean's standard three (propext, Classical.choice, Quot.sound).

Result	Lean
Theorem 2.2	`event_retorsion`, `discrimination`
Definition 3.1	`World`, `Claim`, `Undeniable`, ...
Proposition 3.3	`undeniable_trivial`
Proposition 3.4	`freeze`, `two_models`
Theorem 3.5	`obstruction`, `still_never_runs`
Corollary 3.6	`no_undeniable_bridge`
Definition 4.1	`Successive`, `OrientedSuccessive`
Theorem 4.2	`oriented_tick`
Theorem 4.3	`oriented_chain_runs`
Proposition 5.1	`premise_independent`, `premise_fails_still`, `premise_holds_running`, `no_claim_sett`
Proposition 6.1	`invariant_succ_classified`, `invariant_says_is_ne`
Theorem 6.2	`invariant_succ_symm`, `no_invariant_orientation`
Theorems 8.1, 8.2	`obstruction_tensed`, `horn_two`
Section 7	`lean/Becoming.lean` (`moves_iff_two`, `mark_moves_iff_two`, ...)

Check with `lean lean/Becoming.lean` and `lean lean/Obstruction.lean` against toolchain `leanprover/lean4:v4.31.0`.

11 Conclusion

Discrimination yields occurrence. Under a succession-blind claim language, it does not yield succession: undeniability and running are incompatible once a still world that discriminates the relevant denial is granted — freeze supplies the still structure over a given world, and discrimination at that world is the ostensive step of Section 2. The succession premise P is independent of every claim true at the two-model witness, relative to the retorsive signature, and its orientation clause is invisible to invariant structure. The a priori core is therefore the conditionals together with this limit. Time, if wanted, is paid for separately.

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